

Q6		
(a)(i)	For any given metal, electrons are emitted only when the frequency of incident light is above some minimum value. This frequency is known as the threshold frequency.	
(a)(ii)	At threshold frequency, energy of photon matches the work function. Hence, $\Phi = \frac{hc}{\lambda_o}$ $E_{MAX} = hf - \Phi = \frac{hc}{\lambda} - \frac{hc}{\lambda_o} = hc \left(\frac{1}{\lambda} - \frac{1}{\lambda_o} \right)$	
(b)(i)	Extend line to the x-axis. The x-intercept is $\frac{1}{\lambda_o}$ $\frac{1}{\lambda_o} = 2.3 \times 10^6$ $\lambda_o = 4.35 \times 10^{-7} \text{ m}$	
(b)(ii)	Gradient of graph = hc $\frac{(3.1 - 1.0) \times 10^{-19}}{(3.85 - 2.8) \times 10^6} = h(3.00 \times 10^8)$ $h = 6.7 \times 10^{-34} \text{ Js}$	
(c)	Same gradient Higher y-intercept, x-intercept more to the left.	
(d)	As electrons escape the surface, the sphere becomes more positively charged and the potential at the surface increases. It comes to a point when the electrons with maximum kinetic energy lose all the energy at almost infinite distance and they are attracted back to the surface due to the electric field. The system is finally in dynamic equilibrium. The rate of return is equal to the rate of emission, keeping the total charge constant at the surface. Thus, electric potential energy gained by an electron at infinity = E_{MAX} lost $U_{\infty} - U_{surface} = E_{MAX}$ $0 - \frac{Q(-e)}{4\pi\epsilon_o r} = hc \left(\frac{1}{\lambda} - \frac{1}{\lambda_o} \right)$ $Q = \frac{4hc\pi\epsilon_o r}{e} \left(\frac{1}{\lambda} - \frac{1}{\lambda_o} \right)$	

